

INTERNSHIP REPORT
ON
SMART ACCESS GAUGE(SAG)

Performed at
WIKA Instruments India Pvt. Ltd.

By
SHANMUGA PRIYA R

Duration:
(JAN 2026 – APRIL 2026)



SASTRA
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THANJAVUR, TAMIL NADU, INDIA – 613 401

Year of submission 2026



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Dr. V. Badrinath
Dean, Corporate Relations & Extension Activities

Lr. No.1105/T&P/dt: 03.12.2025

To
The General Manager
WIKA Instruments India Pvt Ltd
Plot No. 128, Sidco North Phase
Ambattur Industrial Estate, Chennai - 600058

Bonafide Certificate

Sir / Madam,

The below mentioned candidate is our Bonafide student of our University. She is interested to do an Internship in your esteemed organization. I request you to kindly permit to perform her Internship since it forms a part of the academic work. We have also instructed the student to follow the rules and regulations of the organization. She is in the 7th Semester of 4 years B. Tech Degree programme.

Name of the student	Reg. No.	Branch of study	Year
Shanmuga Priya R	126004246	Electronics and Communication Engineering	4 th Year

Duration of an Internship: January 2026 to May 2026

Thank you very much

Yours faithfully

Dr. V. Badrinath
Dean, Corporate Relations & Extension Activities

Dr. V. Badrinath
City Union Bank Chair Professor
Dean, Corporate Relations & Extension Activities
SASTRA Deemed University
Thanjavur - 613 401, Tamilnadu, India.

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Finally, I would like to thank my mentors and team members for their guidance, technical support, and encouragement throughout the project.

ABSTRACT

The **Smart Access Gauge (SAG)** is an embedded system designed for real-time measurement and display of angular position using magnetic sensing.

The system integrates a Hall Effect sensor, and a TFT display (ESP32-C3 microcontroller embedded) to provide accurate and continuous angle monitoring.

The primary objective of this project is to develop a reliable industrial-grade measurement system that can be used in process instrumentation environments.

The system captures magnetic field variations using a Hall sensor, converts them into angular values, processes the data using embedded algorithms, and displays the output in real time.

Key features of the system include:

- Real-time angle measurement
- Calibration and linearization
- Embedded signal processing
- Compact and user-friendly interface

The project demonstrates the practical implementation of embedded systems in industrial instrumentation and overcomes challenges such as noise handling, linearity error, and accurate sensor selection .

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1 Introduction

The **Smart Access Gauge (SAG)** is an embedded system developed for real-time measurement of angular position using magnetic sensing. In industrial environments, accurate and reliable measurement of position or angle is essential for monitoring and control applications.

Traditional mechanical gauges have limitations such as parallax error and lack of digital output. To overcome these issues, the SAG system uses modern embedded technology combining a microcontroller, sensor, and display.

This project focuses on designing a compact, accurate, and user-friendly system that measures angular displacement and displays it in real time. The system is built using an ESP32-C3 microcontroller embedded TFT display, a Hall Effect sensor, making it suitable for industrial applications.

2 Smart Access Gauge (SAG)

2.1 Overview

The Smart Access Gauge is designed to measure angular displacement by detecting magnetic field variations. A magnet is attached to a rotating shaft, and a Hall Effect sensor detects changes in the magnetic field as the shaft rotates.

The sensed data is processed by the ESP32 microcontroller and converted into angular values. These values are then displayed on a TFT screen in real time.

The system is compact, efficient, and capable of providing continuous monitoring.

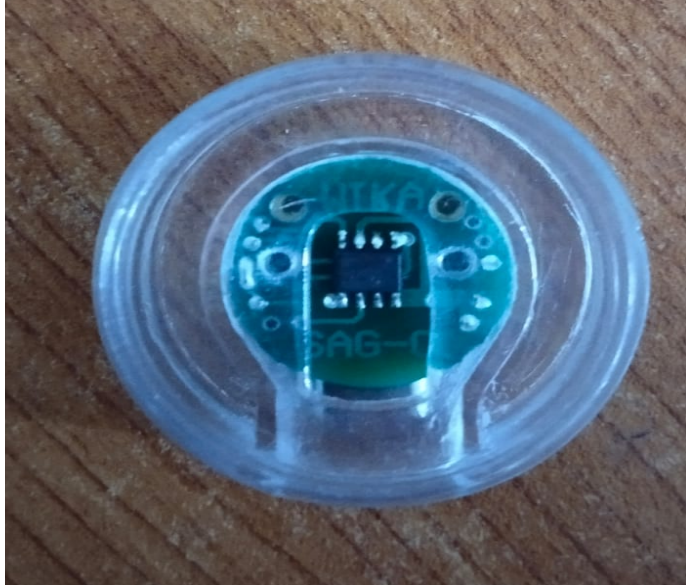


2.2 Working Principle

The working of SAG is based on the Hall Effect principle.

- A **magnet is attached to a rotating pointer**
- A **Hall Effect sensor** detects magnetic field variation
- The sensor outputs a signal corresponding to position
- The **ESP32 processes the signal**
- Angle is calculated and displayed on TFT





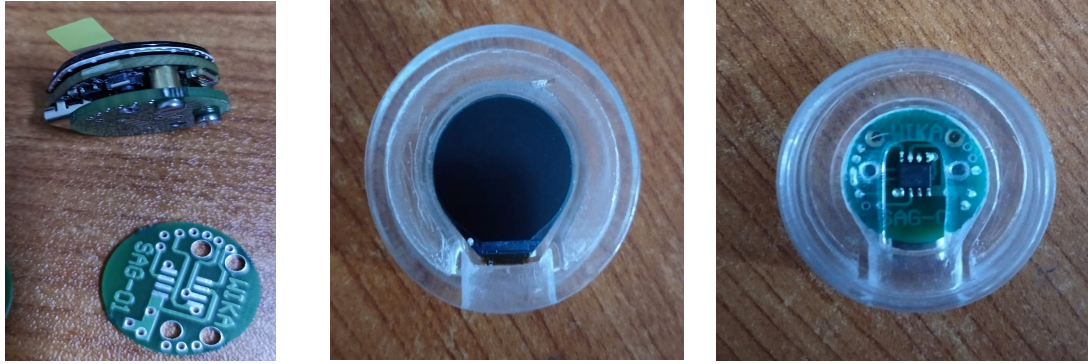
3 Design and Implementation

3.1 Hardware

The hardware consists of the following components:

- **Hall Effect Sensor** – detects magnetic field
- **Display embedded with ESP32-C3 Mc** – displays Pressure
- **PCB** – integrates Mlx90363 to display
- **C cable**– provides required voltage and data transfer
- **3D Printed Enclosure** – holds system components

The PCB was designed to ensure compactness and proper connectivity between all components. The mechanical structure was developed using 3D printing to securely hold the pcb and display.



3.2 Software

The software for the Smart Access Gauge was developed using Embedded C++ in the **Arduino IDE** environment on ESP32-C3. The program is designed to handle sensor communication, data processing, user interaction, and display output efficiently.

Main Functionalities:

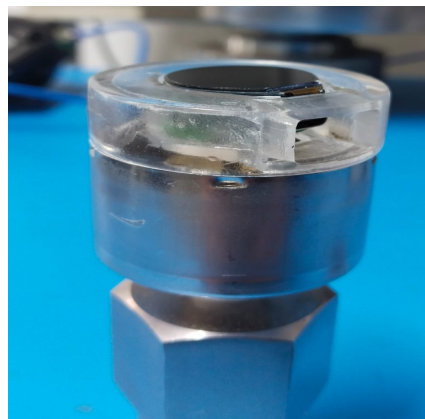
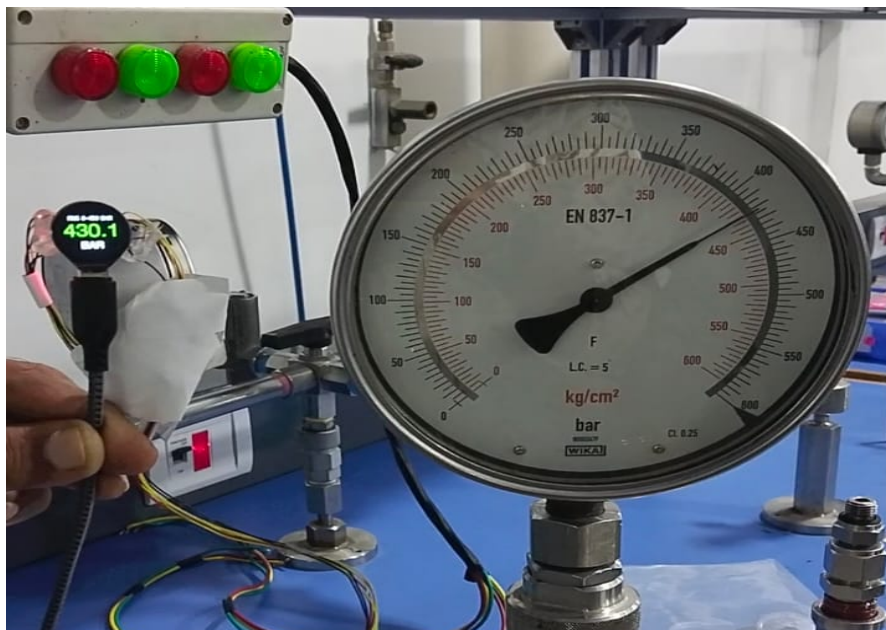
- Reading angular position data from the magnetic sensor using SPI communication
- Converting angle values into pressure using calibration data
- Supporting calibration (LOW and HIGH points)
- Allowing selection of pressure ranges (0–35 BAR / 0–450 BAR)
- Providing gas selection options (Oxygen, Nitrogen, Argon, Neon)
- Storing calibration and configuration data using non-volatile memory (Preferences)
- Displaying real-time pressure values on TFT screen

Algorithm:

1. Initialize display, SPI communication, and memory
2. Read angle data from sensor
3. Apply calibration (angleMin and angleMax)
4. Convert angle to pressure using linear mapping
5. Check selected range and gas type

6. Display pressure value on TFT
7. Update display only when significant change is detected

The software also implements a state-based system (IDLE, CALIBRATION, RUN, etc.) to handle different modes of operation efficiently. Calibrated data is stored in memory, ensuring that the system retains values even after restart. The program is optimized to provide stable output by updating the display only when noticeable changes occur, reducing flickering and noise effects.





4 Testing & Result

The Smart Access Gauge system was successfully developed and tested, achieving real-time pressure measurement with stable and accurate output. Calibration improved the system's accuracy, and the device responded quickly to changes in input. Overall, the system demonstrated reliable performance during testing.

5 Conclusion

The Smart Access Gauge prototype successfully demonstrates integration with a mechanical gauge. The implementation achieves real-time pressure visualization with reliable calibration and effective user interaction.